Review

Blockchain-Assisted Mission Trust in Air–Surface–Underwater Unmanned Systems: A Cross-Layer Review and Object–Assurance–Lifecycle Framework

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Abstract

Air–surface–underwater missions combine unmanned aerial vehicles, unmanned surface vehicles, and unmanned underwater vehicles across heterogeneous links, intermittent connectivity, and multiple administrative domains. Security studies usually examine one platform, network, or ledger mechanism at a time, leaving a gap between trustworthy observations and defensible mission decisions. This narrative review synthesizes literature on cross-domain cooperation, unmanned-system security, trust and reputation, provenance, remote attestation, zero trust, and permissioned blockchains. It proposes the Object–Assurance–Lifecycle (OAL) framework, which relates four trust objects (entity, platform/device, link/path, and data product) to four assurance layers (observation validity, commitment and provenance integrity, ledger and governance consistency, and mission-use suitability) across six mission stages. An object-dependency graph and a context–temporal–graph reasoning loop explain how evidence and attacks propagate while preserving uncertainty about missing or degraded evidence. The review locates distributed-ledger value in coordinating membership, commitments, versions, audit, and post-partition reconciliation, while assigning physical truth, endpoint security, partition-aware safety, hard real-time control, and governance to complementary mechanisms. An evidence-maturity ladder and a proposed Cross-Domain Unmanned Trust Profile turn the synthesis into a testable research and standardization agenda.

Keywords: mission trust; unmanned systems; UAV–USV–UUV cooperation; blockchain; data provenance; remote attestation; zero trust; maritime autonomy

1. Introduction

Unmanned maritime operations are becoming systems of systems. A UAV can rapidly survey a wide area and maintain a radio link; a USV can loiter, carry compute and communications equipment, and bridge air and underwater segments; a UUV can collect close-range observations where radio and satellite services do not reach. Cross-domain control studies increasingly coordinate such platforms for search, tracking, formation, navigation, and surveillance missions [1–7]. The complementary capabilities are operationally attractive, but they also couple domains whose communication physics, failure modes, owners, and security assumptions differ substantially. Broad reviews of UAV networks and applications emphasize mobility, limited energy, intermittent links, and safety constraints [8–10]; reviews of unmanned marine and underwater vehicles add acoustic delay, difficult localization, harsh deployment conditions, and constrained recovery [11–13]. A mission

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decision may therefore depend on evidence that has crossed several media, gateways, processing stages, and administrative boundaries.

This creates a *mission-trust gap*. Conventional cybersecurity questions—who authenticated, which packet arrived, whether a file hash matches, or whether replicas agree—each characterize a specific assurance property; mission-use suitability requires these properties to be composed for a particular decision at a particular time. A correctly signed sonar report can be physically implausible; a valid observation can be replayed after it becomes stale; a correctly ordered ledger entry can originate from a captured endpoint; and a well-provenanced product can still be unsuitable for a high-consequence task because its uncertainty is too large. Research on UAV and swarm security catalogs spoofing, jamming, routing attacks, capture, malware, data manipulation, and privacy threats [14–17]. Underwater-network security reviews describe related attacks under low bandwidth, long propagation delay, and expensive retransmission [18–21]. These literatures provide the mechanisms from which an end-to-end chain from physical observation to mission use can be constructed.

Blockchain is frequently proposed as an answer to distributed trust. UAV-oriented reviews report uses in identity, authentication, coordination, access control, data sharing, and audit [22–26]. Representative proposals use ledgers for swarm identity, lightweight authentication, trusted message exchange, and distributed data acquisition [27–31]. Maritime reviews similarly associate blockchain with documentation, information sharing, authentication, and inter-organizational coordination [32–34]. The attraction is understandable: several organizations can maintain a shared, append-oriented record without granting one participant unilateral control over history. Precise use of the word *trust* distinguishes replicated-state consistency from sensor truth, endpoint integrity, privacy, and timeliness. General blockchain security and Internet-of-Things (IoT) surveys document the associated scalability, resource, governance, privacy, and interoperability conditions [35–38]. The resulting analytical task is to map ledger functions to the claims they support, the complementary controls they require, and the deployment conditions under which shared state adds value.

Adjacent research offers parts of the answer. Zero trust rejects implicit confidence based only on network location and instead emphasizes explicit, least-privilege, continuously evaluated access [39,40]. Remote attestation separates the production of device evidence, its appraisal, and the relying party's decision [41–43]. Provenance models describe how entities, activities, and agents are related and can support accountability across transformations [44–47]. Trust-management research supplies reputation, probabilistic, subjective-logic, fuzzy, and learning-based methods [48–53]. Yet these mechanisms are normally discussed in separate architectural vocabularies. A cross-domain mission requires them to be aligned around the objects being judged, the assurance claim being made, and the stage at which evidence is needed.

This review advances one central argument: *mission trust is a chain of non-substitutable assurance claims, with blockchain strengthening selected transitions within that chain*. To make the argument operational, we propose the authors' *Object–Assurance–Lifecycle* (OAL) framework. It relates four trust objects—entity, platform/device, link/path, and data product—to four assurance layers—observation validity, commitment and provenance integrity, ledger and governance consistency, and mission-use suitability—over six lifecycle stages from admission to recovery. A cross-cutting assurance and governance plane coordinates identities, attestation results, provenance commitments, trust state, policy and model versions, ledger state, and consortium rules across the four mission objects.

The review makes five further contributions. First, it represents dependencies through a typed object graph with relations such as *controls*, *hosts*, *observes*, *transmits*, *transforms*, *endorses*, *records*, and *consumes*. This graph joins evidence propagation to attack propagation. Second, it formulates dynamic trust as a context–temporal–graph (CTG) reasoning loop in which evidence is appraised, attributed, propagated with uncertainty, converted into a policy action, audited, and fed back. Third, it introduces a unified threat-analysis sentence—attacker, capability, object, propagation path, mission impact, and control—while explicitly separating malice, benign environmental degradation, and insufficient evidence. Fourth, it supplies an evidence-maturity ladder from conceptual argument to sustained deployment. Fifth, it proposes Cross-Domain Unmanned Trust Profile (CDUTP) as a forward-looking interoperability profile that maps mission-trust evidence to existing zero-trust, attestation, provenance, communications, and permissioned-ledger specifications.

The analysis focuses on civil applications and publicly documented defense-support settings, where coalition ownership, contested communication, and mission criticality make governance especially consequential. This public-evidence scope supports reproducible analysis of assurance architecture, evidence semantics, failure modes, and evaluation methods. The contribution is a conceptual integration, a set of technology-selection boundaries, and an evaluation agenda for future empirical and formal studies.

The remainder of the paper describes the review scope and analytical lens (Section 2), establishes the operational and conceptual foundations, compares complementary research landscapes, develops OAL and CTG reasoning, examines blockchain patterns and boundaries, organizes evaluation evidence by maturity, and closes with research propositions, limitations, and conclusions.

2. Review Scope and Analytical Lens

This article uses a narrative, problem-driven synthesis focused on the conceptual integration of mission trust. The principal evidence window covers English-language publications from 2016 to 6 August 2026, supplemented by earlier foundational work on reputation, distributed-system limits, provenance, and consensus. Peer-reviewed original studies, authoritative reviews, standards, and official technical specifications form the evidential core; clearly identified preprints provide context for frontier developments. Searches were organized around six intersecting themes: UAV blockchain and swarm security; maritime and underwater communication and security; heterogeneous UAV–USV–UUV cooperation; IoT trust and reputation; provenance, attestation, and zero trust; and permissioned-ledger architecture and governance. Seed reviews were extended through topical searches and forward/backward citation tracing. This design supports structured conceptual comparison across literatures that use different units of analysis and evaluation conventions.

The analytical unit is a *claim–object–context* tuple. For each source, we ask: What object is being judged? Which assurance property is supported? At what mission stage and under what network, threat, and governance assumptions? What evidence backs the claim, and how mature is that evidence? A supplementary matrix records bibliographic identity, source type, domain, platform, mission, trust object, assurance layer, lifecycle stage, evidence, threat, trust method, blockchain role, assumptions, evaluation maturity, and limitations. Bibliographic metadata and DOI resolution were checked independently. Detailed method, assumption, and limitation comparisons draw on records completed through lawful full-text inspection; metadata-level and screened records provide discovery and thematic orientation. The granularity of each claim is thereby aligned with the granularity of its supporting evidence.

The final evidence refresh, completed on 31 August 2026, obtained lawful full texts for all 33 sources in the priority acquisition set. Sixteen sources received claim-level coding for the method-specific comparisons developed below; the remaining acquired sources were screened and retained as a transparent roadmap for subsequent coding. Acquisition and analytical coding are recorded as separate workflow stages, and detailed claims link to completed claim-level records.

The synthesis proceeds through conceptual comparison. Convergent findings define stable design pressures, such as constrained underwater links or the resource cost of on-device ledgers. Divergent findings are treated as conditional: for example, a consensus mechanism that is acceptable on shore servers may be unsuitable on a battery-limited UUV, and a trust threshold valid for routine monitoring may be unsafe for emergency intervention. Statements such as "the literature emphasizes" are attached to explicit groups of sources. The proposed OAL framework, dependency grammar, CTG loop, maturity ladder, and CDUTP constitute the authors' synthesis of this evidence and define an integrated architecture for future empirical validation.

The evidence base includes both general and domain-specific sources. General IoT and blockchain literature supplies reusable mechanisms, while cross-domain unmanned-system studies supply operational constraints. Cross-domain transfer is treated as an engineering hypothesis and evaluated against the target environment's delay, connectivity, resource, threat, and consequence conditions. For example, a road-vehicle trust model can inform context-aware evidence aggregation when its assumptions are re-evaluated under acoustic delay; an industrial permissioned ledger can inform policy consistency when its reconciliation behavior is tested for disconnected underwater operation.

Table 1 summarizes the analytical focus and its complementary context. Satellite, high-altitude, shore, cloud, and edge nodes serve as supporting infrastructure for the three vehicle domains. The mission lifecycle extends through audit, revocation, and recovery because trustworthy operations require action after anomalies: credentials expire, compromised devices are quarantined, policies roll back, partitions reconcile, and evidence remains available for accountability. This endpoint locates blockchain's strongest plausible value in shared state and traceable decisions across organizations, with hard real-time physical control assigned to local safety mechanisms.

Table 1. Review boundary and interpretive choices.

Dimension	Analytical focus	Complementary context
Physical domains	UAV, USV, UUV/AUV/ROV; buoys and cross-media gateways	Ground-vehicle analogies and platform structural-certification requirements
Infrastructure	Satellite, high-altitude, shore, cloud, edge, and consortium services	Partition-aware operation and explicit infrastructure trust assumptions
Mission settings	Observation, inspection, search and rescue, environmental monitoring, logistics, and publicly documented defense-support contexts	Publicly reportable assurance semantics, failure modes, and evaluation methods
Trust mechanisms	Identity, authorization, attestation, provenance, dynamic inference, policy, ledger, audit, and governance	Validated scoring, protocol verification, and probability calibration as future implementation work
Evidence	English peer-reviewed work, authoritative reviews, standards, official specifications, and identified frontier preprints	Claim granularity matched to verified metadata and completed full-text coding
Review output	Conceptual framework, taxonomy, design boundaries, maturity ladder, and research propositions	Empirical comparison, formal verification, and deployment certification as evaluation stages

3. Operational Context and Trust Foundations

3.1. A Heterogeneous Mission System of Systems

The core system contains UAVs, USVs, and UUVs that contribute different sensing, mobility, computation, and communications functions. UAVs offer rapid repositioning, line-of-sight radio relays, and electro-optical sensing, but their payload, endurance, and link stability are constrained. USVs can carry larger energy stores and antennas, remain on station, and bridge acoustic and radio networks. UUVs provide proximity to underwater targets and environments but frequently depend on low-rate acoustic channels, short-range optical links, or delayed surfacing for high-volume transfer. Surveys of UAV-aided maritime communication and 6G maritime connectivity illustrate the relay opportunities [54,55]; underwater communication reviews document the trade-offs among acoustic, optical, and other media [56–59]. The resulting system requires a dynamic network model in which topology, latency, bandwidth, clock quality, and reachability can change during a single mission.

Full-text inspection sharpens this point. Civil-UAV communication requirements are explicitly mission-dependent: application classes vary in network role, data volume, range, and quality-of-service needs [8]. Maritime-search evidence characterizes heterogeneous cooperative systems as an early-stage applied field built on increasingly mature UAV, USV, and UUV capabilities [60]. Mission trust is therefore conditioned on current role and mission phase, with evidence re-evaluated whenever the application domain or consequence changes.

Cross-domain cooperation also changes fault containment. A USV may be the navigation reference, data cache, radio gateway, and permissioned-ledger peer for several UUVs. Compromise or benign failure of that high-betweenness platform therefore affects link evidence, time, provenance continuity, and access to the shared record. Conversely, platform diversity can improve resilience when evidence from independent media and locations can corroborate an event. Cooperation research demonstrates the control and planning value of heterogeneous formations and multi-swarms [5,6,61,62]. Mission assurance extends these results by making the trustworthiness of shared observations, commands, and dependencies explicit.

Administrative heterogeneity is equally important. A civil emergency may combine government responders, commercial communications providers, research vehicles, and contracted data services. A coalition mission may include platforms that remain under distinct authorities. Cross-organizational authority therefore combines local authentication with policy equivalence, delegated roles, and accountable governance. The system boundary includes organizations, credentials, data-use restrictions, software and model versions, and the process that admits, suspends, or removes participants. Under these conditions, a shared ledger can provide a consistent audit state whose history is jointly governed by the participating parties.

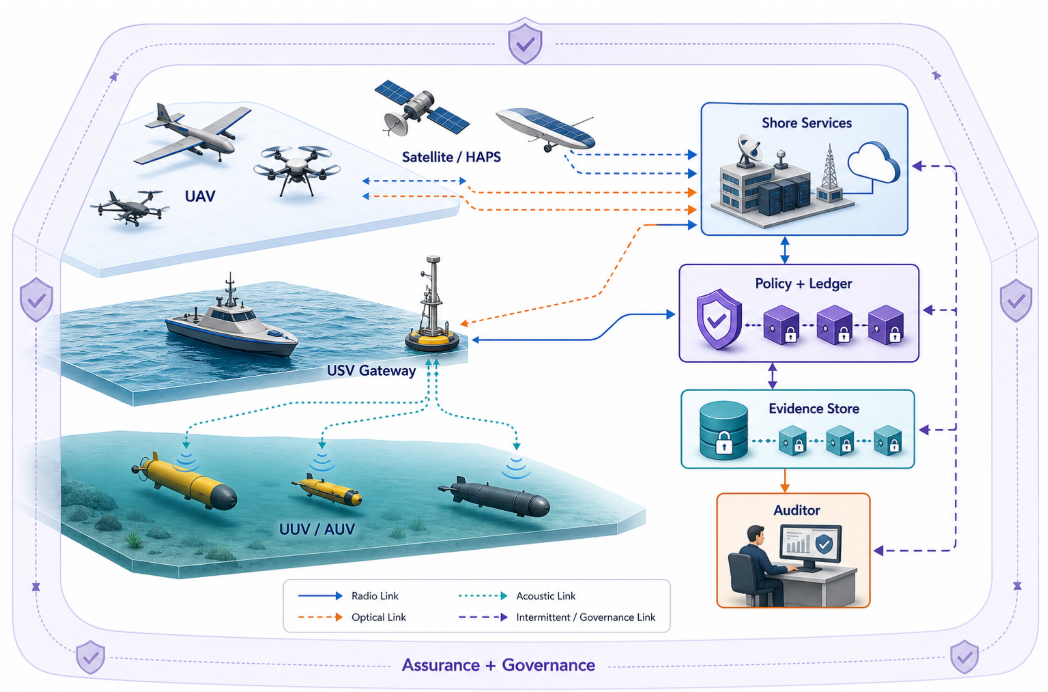

Figure 1. Author-proposed cross-domain reference architecture. Full ledger peers and policy services are placed on relatively capable USV, shore, or edge nodes; constrained vehicles retain signed local logs and submit commitments when connectivity permits. The dashed enclosure identifies the cross-cutting assurance and governance plane.

Figure 1 presents an adaptable logical deployment. Raw imagery, sonar, telemetry, and detailed attestations remain encrypted off chain. A permissioned ledger may retain content identifiers, commitments, coarse-grained state transitions, membership changes, policy and model versions, revocations, and audit decisions. During disconnection, an edge node keeps an append-only signed sequence and later submits a batch commitment. Reconciliation reveals gaps, duplicates, invalid credentials, and conflicting histories; local sensing and control mechanisms retain responsibility for physical interpretation and real-time deadlines.

3.2. Mission Trust, Trustworthiness, and a Trusted Data Environment

We define *mission trust* as a context-specific relying decision about whether an object may contribute to a stated mission purpose, given available evidence and the consequences of error. Trust is therefore relational and prospective: a consumer relies on an object for an action. *Trustworthiness* denotes properties that justify such reliance, including identity authenticity, platform integrity, link availability, provenance, freshness, data quality, and accountable governance. Both vary with time and context. The same UAV may be suitable for low-risk environmental imagery while being excluded from time-critical navigation support because its attestation is stale or its location is uncertain.

For an object o, time t, context c, and mission use m, a general representation is

$$T_o(t, c, m) = \Pr(o \text{ satisfies the properties required for } m \mid \mathcal{E}_{1:t}, c), \quad (1)$$

where $\mathcal{E}_{1:t}$ denotes the direct and indirect evidence currently available. Equation (1) provides a conceptual notation for context-dependent reliance. Implementations may represent the resulting uncertainty through calibrated probabilities, subjective logic, sets, intervals, ordinal states, or decision rules [51–53]. In each representation, uncertainty and evidence lineage remain explicit alongside the trust assessment.

A *trusted data environment* is the socio-technical setting that produces, transports, transforms, shares, and governs evidence and mission data. It includes principals, devices, paths, data products, policies, models, services, and accountability processes. Mission trust is the decision relation exercised within that environment. A tamper-evident catalog strengthens the environment, while mission-use appraisal determines which cataloged item is fit for a particular decision. Zero trust contributes the access-control architecture: NIST's formulation uses policy engines, identity and device evidence, enforcement points, and protected resources [39]. Intermittently connected missions add safe degradation and visible evidence staleness to those functions.

3.3. Four Non-Substitutable Assurance Claims

The assurance chain in Figure 2 distinguishes four questions. Observation validity asks whether a measurement plausibly represents the physical phenomenon, considering calibration, geometry, environment, sensor health, and corroboration. Commitment and provenance integrity asks who or what produced the item, when and where it was produced, how it was transformed, and whether the recorded bytes and lineage were altered. Ledger and governance consistency asks whether authorized record keepers agree on membership, order, versions, and state transitions under stated fault assumptions. Mission-use suitability asks whether the item is timely, relevant, sufficiently certain, policy-compliant, and proportionate to the consequences of the contemplated action.

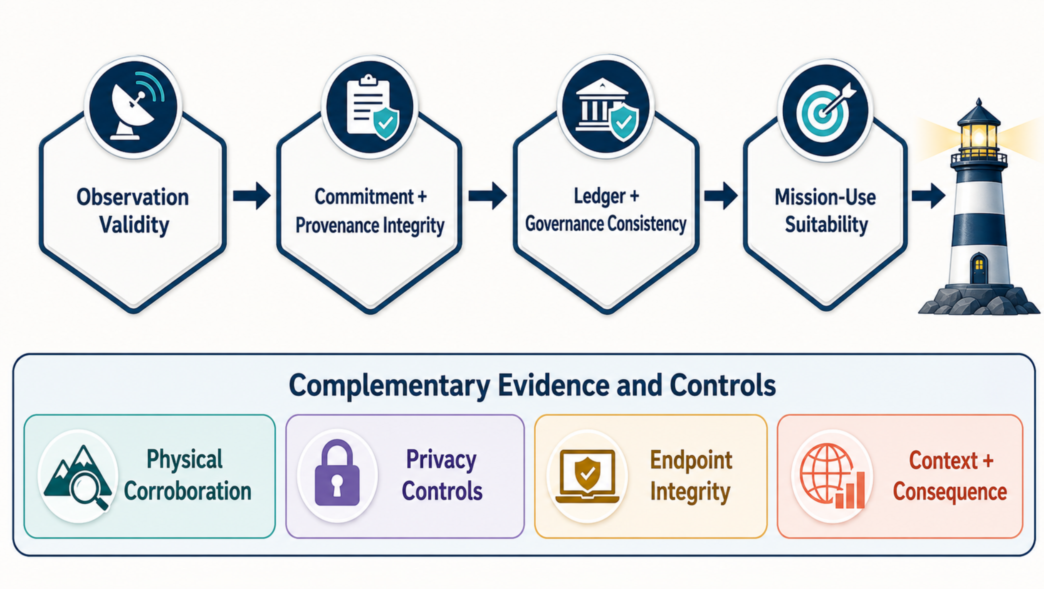

Figure 2. The authors' four-layer assurance chain. Each layer contributes a distinct claim to the relying decision.

These claims have distinct failure modes. GNSS spoofing can make an otherwise healthy UAV report the wrong location; machine-learning and multi-sensor approaches may detect inconsistency, with validity conditioned on data and context [63–67]. A stolen signing key can produce syntactically valid provenance. A permissioned consensus group can faithfully order false submissions. A delayed but accurate observation can be unfit for collision avoidance. The chain therefore keeps four concepts distinct: signatures and physical truth; hashes and privacy; consensus and endpoint security; and historical reputation and current mission fitness.

Remote attestation and device baselines fit between the first two layers. The RATS architecture distinguishes evidence, appraisal policy, attestation results, and relying parties [41]; the NIST IoT baseline emphasizes device identification, configuration, data protection, logical access, software update, and cybersecurity-state awareness [68]. These mechanisms justify scoped claims about measured software and device state through roots of trust, reference values, verifier freshness, and explicit measurement coverage. External observation validity is established through complementary physical and sensor evidence. Provenance records derivation: W3C PROV relates entities, activities, and agents, enabling consumers to ask how a product was generated [44,45]. OAL brings these distinct claims into one mission-oriented vocabulary.

4. Complementary Research Landscapes

The relevant evidence is substantial and distributed across technical communities. Table 2 compares the principal neighboring review families against the dimensions needed for mission trust. A check mark indicates sustained treatment; a circle indicates partial or enabling treatment; a dash identifies a dimension outside the family's central organizing concern. The interpretive comparison shows how the distinct primary questions of each family contribute to a common cross-layer model.

Table 2. Comparison of neighboring review families and the cross-layer gap. Symbols: ✓, sustained coverage; ○, partial/enabling coverage; –, outside the central focus.

Review family	Air/sea/underwater	Four objects	Assurance chain	Dynamic uncertainty	Ledger boundary	Lifecycle	Sources
UAV networks, applications, and security	○	○	–	○	–	○	[8,14–16]
UAV blockchain	–	○	○	–	○	○	[22–25]
Unmanned marine and cross-domain cooperation	✓	–	–	–	–	○	[11,12,54,60]
Underwater communication and security	○	○	–	○	–	○	[18,21,59,69]
IoT trust and reputation	–	○	–	✓	○	○	[48–50,70]
Provenance, attestation, and zero trust	–	○	○	○	–	✓	[40,46,47,71]
General IoT blockchain	–	○	○	–	○	○	[36,37,72,73]
This review and OAL synthesis	✓	✓	✓	✓	✓	✓	Authors' proposed analytical framework

4.1. Cross-Domain Cooperation and Assurance Semantics

Cooperation studies clarify what platforms can do together. They address formation control, cooperative path planning, communication support, target search, and distributed consensus in the control-theoretic sense [2–5,7,62]. These studies make heterogeneity explicit and sometimes model disturbances or link constraints. In this literature, *consensus* usually denotes agreement of agent states or coordinated control, whereas replicated-ledger consensus concerns shared record state. OAL connects the coordination output to security provenance, identity delegation, and cross-organizational audit by treating it as a data product with traceable inputs, transformation, policy version, and mission use.

Maritime search and unmanned-vehicle reviews show that mission effectiveness relies on distributed sensing, localization, planning, and human oversight [11,13,60]. These dependencies motivate claim-specific evidence across platform, path, and data-product states. A UUV may be healthy while its acoustic path is jammed; a USV gateway may authenticate yet relay a stale model; a UAV may provide valid imagery whose georegistration is wrong. The mission consumer can therefore base each relying decision on the relevant combination of claims instead of one platform-wide label.

4.2. From Security Catalogs to Cross-Object Propagation

UAV, swarm, and underwater-security reviews offer mature attack taxonomies. Typical categories include physical capture, identity and authentication attacks, GNSS spoofing, jamming, denial of service, malicious routing, eavesdropping, malware, false-data injection, and privacy leakage [14,15,17–19]. Hierarchical detection and response illustrates how monitoring and countermeasures can be distributed in a UAV network [74]. Dependency analysis extends these catalogs by tracing how an initial capability traverses objects. When a captured gateway hosts a route, transforms observations, endorses a batch, and records a commitment, effective controls span the physical endpoint, semantic transformation, and ledger interface.

Environmental degradation creates an additional classification problem. Acoustic fading, multipath, Doppler, marine noise, weather, low energy, and topology change can produce loss, latency, or disagreement that resembles attack. Underwater-network reviews repeatedly emphasize these physical constraints [56,57,75,76]. A trust system that equates anomalies with malice may isolate the very nodes needed during severe conditions. Conversely, attributing all anomalies to the environment invites an adversary to hide within expected variation. A cross-object dependency model must preserve at least three hypotheses: malicious manipulation, benign degradation, and insufficient evidence.

4.3. Trust Computation with Mission Semantics

IoT trust and reputation literature provides broad taxonomies of direct observation, recommendation, context, centralized and distributed management, and attacks on reputation [48–50,77]. Underwater studies have explored decision-tree and hidden-Markov approaches [78,79], and design guidance recognizes underwater-specific evidence constraints [80]. These methods show how a belief can be updated. OAL supplies its mission semantics by requiring each estimate—for example, honest routing, current software integrity, sensor accuracy, or willingness to cooperate—to name the object, property, context, evidence lineage, uncertainty expression, and permitted use.

Blockchain-based trust management promises shared reputation and tamper-evident updates across IoT participants [70,81–83]. The ledger reduces unilateral history rewriting and makes the quality of the underlying trust model consequential over a longer history. Colluding recommendations, correlated observations, model drift, and incorrect identities are addressed through upstream evidence controls. Storage and governance choices therefore follow the definition of evidence semantics.

Fog and edge trust approaches demonstrate the attraction of placing appraisal close to constrained devices, while broader trust-recommendation reviews identify context, recommendation quality, and attack resistance as continuing design concerns [84,85]. In a cross-domain mission, a nearby USV edge service may observe several UUVs while remaining their common gateway. A local estimator reduces delay, and the dependency graph records the platform, clock, path, and administrative owner shared by its recommendations.

4.4. Ledger Mechanisms and Physical-Evidence Boundaries

General blockchain reviews explain architectures, consensus mechanisms, smart contracts, scalability, and security [86–90]. IoT integrations confront limited computation, storage, communication, energy, privacy, and interoperability [36,37,91]. UAV implementations illustrate authentication, clustering, data acquisition, and swarm coordination [31,92–94]; maritime implementations show authentication and signcryption around shared logistics infrastructure [95,96]. These design precedents define concrete choices for membership, key custody, endorsement, validation, and the binding between on-chain commitments and off-chain content.

Cross-sector classifications show that blockchain applications inherit recurring problems of scalability, data quality, integration, and governance [97]. UAV-specific taxonomies and 6G perspectives reach a similar conclusion from the networking side: ledger placement is conditioned by resource, mobility, and software-defined-network constraints [98–100]. These observations support selective deployment on capable gateways and edge nodes. OAL supplies the additional criterion for selecting the mission assurance claims that benefit from ledger support.

The synthesis opportunity is a common analytical model connecting a physical observation, the responsible principals and devices, the paths and transformations, the shared governance record, and the eventual mission decision. The next section develops that model.

5. The OAL Framework for Mission Trust

The authors' *Object–Assurance–Lifecycle* (OAL) framework is a three-dimensional classification and design instrument. It can be used to locate a published mechanism, specify a required assurance case, or identify an unprotected transition. The dimensions are intentionally orthogonal. An object says *what is judged*; an assurance layer says *which claim is sought*; and a lifecycle stage says *when the claim is needed*. A mechanism may occupy several cells while preserving the identity of each axis. Figure 3 presents the whole framework.

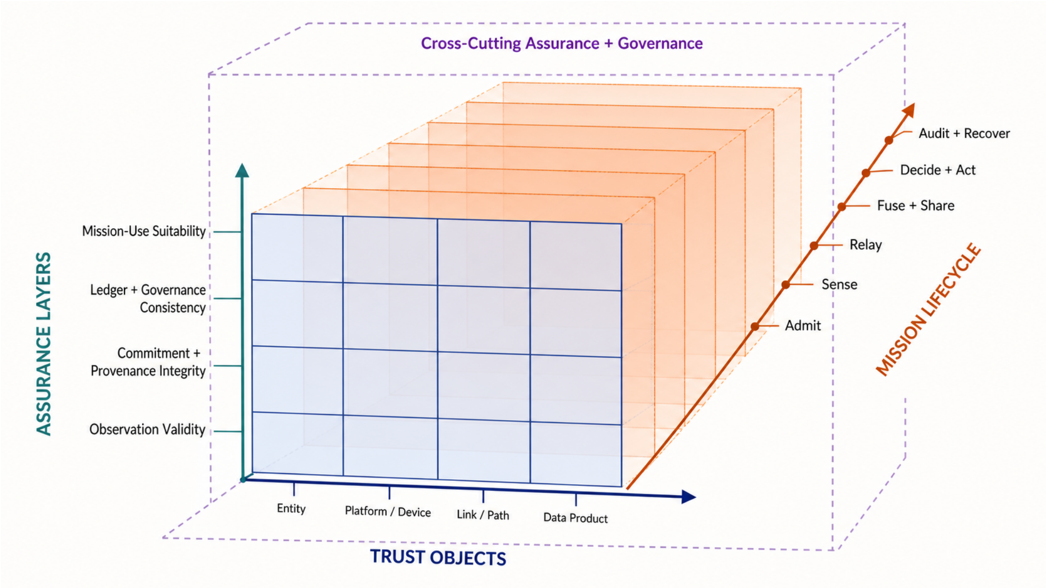

Figure 3. The authors' OAL framework. Trust objects run along the horizontal front-face axis, assurance layers along the vertical axis, and lifecycle stages along the oblique depth axis; together they define independent cells within the cross-cutting assurance and governance plane.

5.1. Dimension I: Four Trust Objects

An *entity* is a principal capable of holding authority or responsibility: an operator, organization, mission service, software agent, or autonomous decision-making principal. A *platform/device* is an executing physical or virtual endpoint: vehicle, sensor, gateway, compute module, secure element, software image, or hosted model instance. This distinction matters because an authorized entity can control a compromised device, while a device with valid attestation can execute a legitimate but unsuitable policy. An autonomous vehicle may occupy both categories under different claims: as a delegated principal for action and as an endpoint whose hardware and software state are measured.

A *link/path* is a time-varying end-to-end route composed of its physical media, intermediate nodes, security associations, and quality state. A data product may traverse a UUV acoustic hop, a USV gateway, a satellite backhaul, and a cloud broker; confidentiality, freshness, availability, and routing integrity depend on this composed path. A *data product* is an immutable version of raw observation, feature set, track, map, fused estimate, model output, command, alert, provenance record, or audit artifact. Treating versions as immutable avoids silent overwriting and permits provenance to express transformation.

Table 3. Four OAL trust objects, evidence, failure examples, and mission questions.

Object	Representative evidence	Illustrative failure or attack	Mission question
Entity	Credential chain, organizational role, delegation, authorization, prior accountable actions	Credential theft, insider misuse, collusion, obsolete authority	Who requests, endorses, or consumes the resource, and under which authority?
Platform/device	Secure boot measurement, firmware/software version, sensor self-test, configuration, energy and health state	Capture, malware, unsafe configuration, stale reference value, sensor drift	Which endpoint executes, and is its measured state adequate for this use?
Link/path	Authentication state, route, delay, loss, signal indicators, neighbor changes, medium and gateway state	Jamming, wormhole, selective forwarding, eavesdropping, benign fading	Through which changing path did evidence pass, and what security and quality properties held?
Data product	Signature, timestamp, location, provenance graph, quality metadata, version, calibration and cross-source residuals	False injection, replay, poisoning, misregistration, unauthorized transformation	Where did this version come from, was it altered, and is it timely and suitable?

Policies, models, smart contracts, and ledger state map onto the four operational object categories. They are versioned artifacts hosted on devices or represented as data products, while the principals who authorize them remain entities. This stable object set accommodates new mechanisms, and the assurance plane makes their artifacts first-class subjects of versioning, access control, threat analysis, and audit.

5.2. Dimension II: Four Assurance Layers

The first layer, *observation validity*, asks whether evidence about the external world is credible. Relevant controls include calibration records, sensor diagnostics, geometry, physical bounds, environmental models, independent modalities, and temporal continuity. Multi-view and multi-sensor fusion can expose disagreement and express uncertainty [101–103]; it can also amplify correlated error or poisoned inputs. Observation assurance therefore documents independence assumptions and combines physical and domain evidence with cryptographic content commitments.

The second layer, *commitment and provenance integrity*, binds an item to a producer, time, place, transformation, and immutable version. Digital signatures, authenticated channels, secure timestamps, content identifiers, and W3C PROV relations support this layer [44–47, 104]. Provenance covers both raw acquisition and model, human, or automated decisions. Decision provenance connects inputs, policy versions, intermediate transformations, and outputs, making later review possible. Its assurance conditions include key custody, clock assumptions, instrumentation completeness, and trustworthy capture of events.

The third layer, *ledger and governance consistency*, concerns a shared state among administrative parties. It covers member admission, endorsement, record ordering, validation, version coordination, revocation, dispute handling, and reconciliation. Hyperledger Fabric illustrates an execute–order–validate architecture with identity-aware channels and endorsement policies [105,106]; other permissioned designs can satisfy similar functions. This layer establishes authorized agreement under a declared fault model, while semantic validity remains grounded in observation and mission-use evidence.

The fourth layer, *mission-use suitability*, converts assurance evidence into a bounded relying decision. Freshness, relevance, uncertainty, risk class, legal and organizational policy, available alternatives, and consequence severity are considered. A consumer may accept a low-resolution image for broad situational awareness but require corroborated, current, and accurately georegistered evidence for autonomous maneuver. Suitability is therefore expressed as a purpose-specific authorization or confidence condition.

5.3. Dimension III: Six Lifecycle Stages

The lifecycle dimension prevents security from ending at admission. Table 4 shows how objects and assurance questions evolve from credentialing to recovery. Evidence can age, dependencies can change, and an initially valid decision can require reassessment.

Table 4. Mission lifecycle and representative OAL assurance tasks.

Stage	Primary questions	Evidence and controls	Ledger-relevant state
Admission and authentication	Is the entity authorized, and is the joining endpoint in an acceptable measured state?	Credentials, delegation, attestation, configuration baseline, least privilege	Membership, role, attestation-result commitment, policy version
Sensing and collection	Does an observation plausibly reflect its stated time, place, modality, and calibration?	Sensor self-test, secure time/location, quality metadata, corroboration	Acquisition commitment and data identifier
Transmission and relay	Which path carried the item, and did quality or security conditions change?	Mutual authentication, route and link telemetry, delay/loss context, custody transfer	Signed relay events or batch roots
Processing, fusion, and sharing	Which inputs, code, model, parameters, and policies produced this version?	Provenance graph, reproducible transformation identifier, access and usage control	Version, endorsement, access decision, model/policy commitment
Decision and action	Is the product fit for the specific mission consequence now?	Uncertainty, freshness, risk policy, alternatives, human review or fail-safe	Decision rationale commitment and responsible principal
Audit, revocation, and recovery	What failed, who was affected, what must be invalidated, and how is operation restored?	Incident evidence, credential and model revocation, rollback, re-attestation, after-action review	Revocation state, dispute outcome, reconciled log, recovery version

Lifecycle continuity is especially difficult during partitions. If a UUV cannot contact a central policy engine, it still needs a local authorization envelope: named resources, permitted actions, risk limit, evidence freshness, expiry, and fail-safe behavior. Local actions are signed and sequenced; reconnecting nodes later disclose commitments and selected evidence. The design distinguishes *continued operation under delegated authority* from *fresh global authorization*; the ledger makes that distinction auditable, and local policy enforces it during disconnection.

5.4. The Assurance and Governance Plane

The cross-cutting plane contains identity and authorization, remote attestation, provenance capture, trust inference, policy/model/contract version management, ledger state, and consortium governance. Each component is itself attackable. Identity authorities can be compromised; reference values can be wrong; provenance instrumentation can omit transformations; a trust model can drift or be poisoned; a smart contract can encode an unsafe rule; validators can collude; and governance can deadlock. The plane must therefore be observed through the same OAL lens. For example, a model is a data product with provenance and validation evidence, a deployed instance is hosted on a device, its approval is an entity action, and its use is constrained by lifecycle stage and mission context.

OAL serves as both a diagnostic and a specification template. A blockchain-throughput study maps to ledger-performance cells and prompts complementary observation, endpoint, and mission-use requirements. A firmware-attestation protocol maps to a device claim at admission and prompts a freshness rule for disconnected decision time. A mature mission-assurance design identifies every high-consequence dependency path, states the claim at each cell, and exposes absent or stale evidence.

6. Evidence Dependencies, Threat Propagation, and Dynamic Reasoning

6.1. A Typed Object-Dependency Graph

OAL classifies claims; a dependency graph explains how they interact. Let $G_t = (V_t, E_t)$ be a time-indexed directed multigraph whose vertices are versioned instances of the four object types. Edges carry typed relations: an entity *controls* a platform; a platform *hosts* a sensor, model, or gateway; a platform *observes* a data product; a link *transmits* it; an activity *transforms* inputs into a new product; an entity or service *endorses* it; a ledger transaction *records* a commitment; and a mission function *consumes* it. Time and version qualifiers are essential. "UAV-7 hosts model M" is insufficient if the decision depended on model version 4.2 before a rollback.

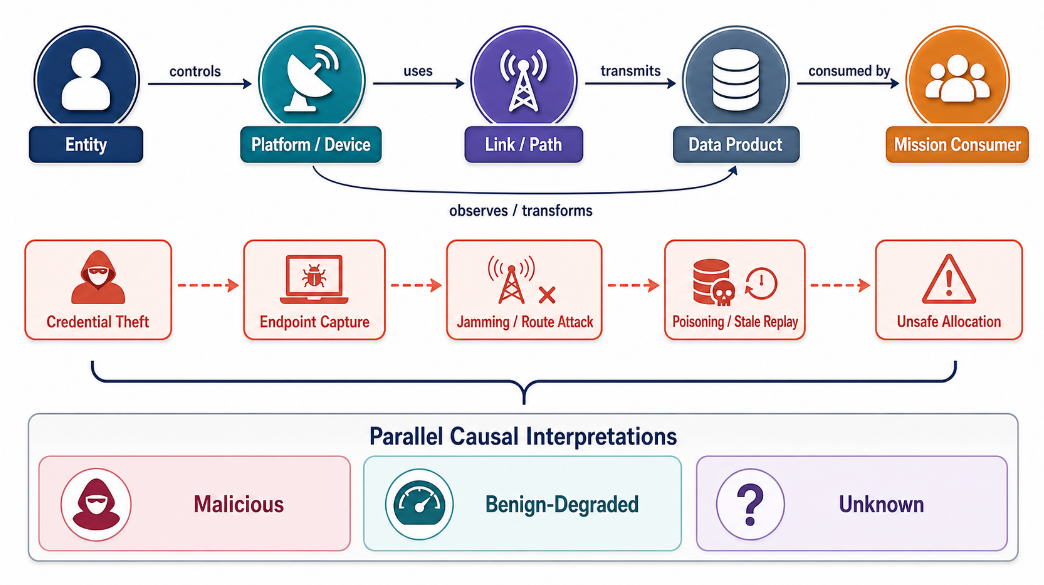

Figure 4. Simplified object dependency and attack-propagation path. Red dashed arrows illustrate one causal hypothesis within a diagnosis that retains malicious, benign-degraded, and unknown interpretations.

The graph unifies evidence and threat reasoning. Evidence can support a local claim and propagate conditionally: a fresh attestation supports the integrity of the platform that generated a product; verified provenance supports the relation between an input and transformation; independent observations support physical plausibility. An attack can traverse the same relations in the opposite epistemic direction. Stolen entity credentials authorize access to a device, device compromise corrupts observations or gateway functions, a manipulated path delays or substitutes a product, and a mission consumer acts on the result. Figure 4 gives one illustrative chain. Propagation models retain correlation, evidence coverage, and non-monotonic path effects instead of reducing the graph to multiplication of trust scores.

6.2. A Unified Threat-Analysis Sentence

Every scenario should be expressible as: *an attacker with capability C acts on object O, propagates through relations P, creates mission impact I, and is constrained or detected by controls K under assumptions A*. This grammar connects threat catalogs to assurance cases and exposes missing controls. Table 5 applies the grammar to common public scenarios at a capability level suitable for reproducible defensive analysis.

Table 5. Object-centered threat, degradation, and evidence-absence analysis.

Class and initial object	Capability or condition	Propagation path	Mission impact	Controls and residual uncertainty
Malicious/entity	Stolen or misused credential	controls device → endorses product → record accepted	Unauthorized tasking or false accountability	Short-lived delegation, multi-party approval, behavioral evidence, rapid revocation; valid credentials can still mask insider intent
Malicious/device	Capture, malware, or model replacement	hosts sensor/model → observes/transforms data	Consistent but false products across many consumers	Attestation, key protection, diversity, provenance, quarantine; measured components may omit sensors or runtime behavior
Malicious/link	Jamming, wormhole, selective forwarding, replay	transmits/custodies data → delays or substitutes version	Loss of availability, stale situational picture, biased fusion	Link context, authenticated sequence, path diversity, freshness policy; attribution remains uncertain
Malicious/data	False-data injection, poisoning, semantic manipulation	transformed / endorsed / recorded product → consumed decision	Incorrect classification, route, or resource allocation	Physical plausibility, independent modality, robust fusion, lineage; independence assumptions can fail
Benign environment	Fading, multipath, weather, occlusion, low energy	degrades link or observation evidence without hostile action	Delay, disagreement, reduced coverage	Environment-aware baselines, graceful degradation, explicit "benign-degraded" hypothesis; models may be incomplete
Evidence absence	Disconnection, failed verifier, missing lineage, stale clock	prevents appraisal across several relations	Unsafe overconfidence or unnecessary isolation	Unknown state, bounded authority, conservative policy, later reconciliation; absence is not proof of compromise

Three-way interpretation provides a baseline state model. *Malicious* denotes evidence supporting intentional or adversarial manipulation. *Benign-degraded* denotes a supported environmental or operational explanation. *Unknown* preserves evidential ambiguity when the available record supports neither conclusion. A recovery state tracks the additional review of cached data, keys, neighbors, and prior decisions after software is re-attested. Policies map these states to bounded actions—continue, reduce privilege, request corroboration, isolate, return, or escalate to a human—as purpose-specific operational conditions.

6.3. The CTG Reasoning Loop

CTG-Trust denotes the authors' context–temporal–graph reasoning loop shown in Figure 5. First, evidence events are captured with source, subject, time, context, value, and lineage. Second, evidence reliability is appraised: a device self-report and an independent verifier carry different evidential relationships. Third, object state is inferred using an appropriate published method. Fourth, dependencies are traversed to identify downstream claims and shared causes. Fifth, uncertainty is made explicit. Sixth, a mission policy chooses an action. Seventh, the decision, relevant versions, and evidence commitment are recorded for accountable review. Eighth, operational outcomes and later verification update the evidence base.

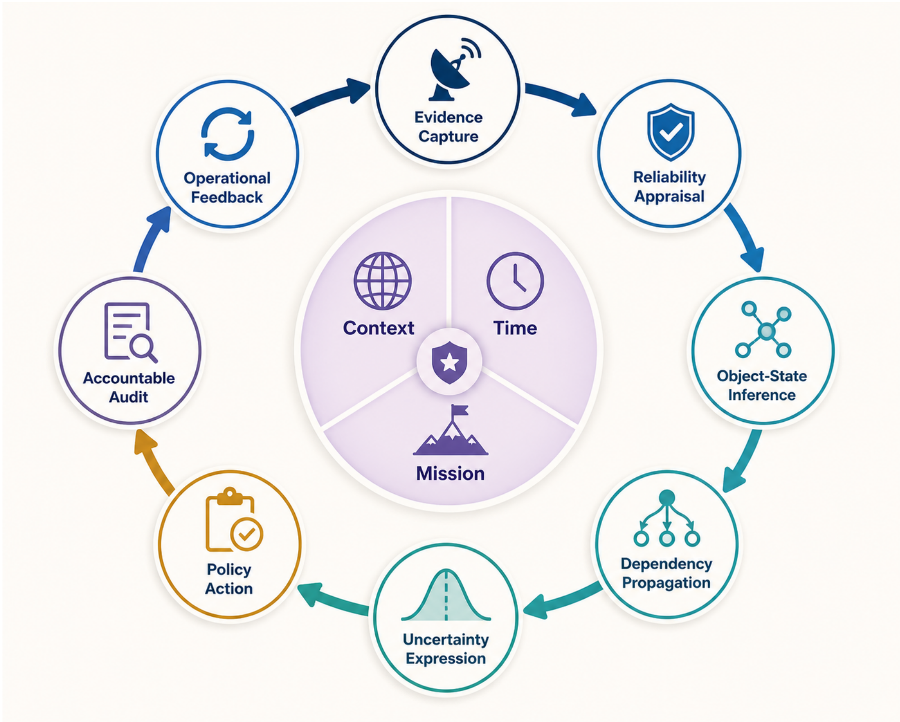

Figure 5. CTG as an author-proposed reasoning loop that defines information flow, uncertainty handling, policy action, and accountability while accommodating validated inference methods.

Evidence reuse requires special care. A gateway's report can influence device reputation, link assessment, data quality, and a final fused decision; treating each derived value as independent counts the same evidence repeatedly. The dependency graph should retain parent identifiers and common causes so that an implementation can discount correlation or avoid duplicate use. Recommendations also carry provenance. Ten pseudonymous recommendations count as independent observations only when membership and organizational independence support that interpretation; ledger persistence preserves the basis for this appraisal.

6.4. Trust Methods and Appropriate Claims

The literature offers several method families, summarized in Table 6. Each expresses a different claim under different assumptions. The qualitative comparison provides a selection guide for future mission-specific validation.

Table 6. Trust-inference families and their role in CTG.

Family	Useful representation	Strength	Critical assumption/risk	Representative sources
Weighted multi-attribute score	Context-weighted evidence features	Interpretable and implementable at constrained edges	Weights and thresholds can be arbitrary or task-insensitive	[48,49]
Bayesian/reputation	Evidence-updated belief or behavior propensity	Natural sequential updating	Priors, stationarity, and evidence independence may be unjustified	[52,53]
Subjective logic	Belief, disbelief, uncertainty, and base rate	Makes ignorance visible and supports opinion combination	Dependence and source identity must be handled explicitly	[51]
State-space/Markov	Latent state with temporal transitions	Represents persistence and recovery	State meanings and transition parameters need mission data	[79]
Rule/tree/fuzzy methods	Human-readable rules or learned partitions	Can combine heterogeneous evidence	Coverage gaps and abrupt boundaries; calibration often unclear	[78,80]
Machine/deep learning	Learned anomaly or multi-view representation	Captures nonlinear patterns	Shift, poisoning, explainability, and overconfidence	[63,64,102]
Graph-based propagation	Relational evidence over dependencies	Connects local evidence to downstream exposure	Correlation, cycles, and collusion can amplify error	[53,81]

The underwater studies illustrate why the right-hand risk column is substantive. The HMM proposal in Arifeen et al. [79] maps packet-loss rate and packet-error rate to latent malicious/trustworthy states; its MATLAB illustration uses ten observations with randomly initialized probabilities and demonstrates a temporal construction under a synthetic numerical setup. TEUC combines data-, link-, and node-based evidence with a C4.5 tree and reward/penalty updates [78]; its software simulation also highlights how acoustic loss can make normal nodes appear untrustworthy and how an unpolluted trust-evidence assumption affects part of the evaluation. The broader UWSN review confirms that evidence collection, evaluation, propagation, and maintenance involve distinct design choices [80]. Together, these results motivate three CTG requirements: keep node behavior distinct from sensor truth, validate attack classification against mission decision loss, and represent benign degradation as an explicit competing hypothesis.

Equation (1) can sit above any of these methods when the score states the represented event, context, horizon, calibration evidence, and decision loss. These fields give numerical precision an interpretable assurance meaning. Beta parameters, hidden-state transitions, scoring thresholds, message schemas, and performance targets then become explicit empirical design choices for each implementation study.

7. Blockchain-Assisted Assurance: Roles and Boundaries

7.1. The Decision to Use a Ledger

The decision to use blockchain begins with the governance structure. A permissioned ledger is plausible when multiple independently administered parties coordinate shared state, history requires joint control, records require accountable endorsement, and intermittent participants reconcile after disconnection. A conventional authenticated database is usually preferable when one accountable organization controls membership and audit, while local verified control serves hard real-time loops. These criteria align consensus cost with a concrete need for cross-organizational state and dispute resolution.

The relevant ledger is normally permissioned. Open, computation-intensive participation is poorly aligned with constrained vehicles, sensitive mission metadata, and accountable membership. Full peers and ordering services belong on capable shore, edge, or selected USV nodes; UAVs and UUVs can act as clients or light verifiers. Hyperledger Fabric provides one representative architecture with identities, endorsement policies, channels, ordering, and validation [105,106]. The OAL framework is ledger-neutral: Byzantine-fault-tolerant replicas, crash-fault-tolerant ordering, or another shared-state service may be selected if the stated threat, availability, and governance assumptions match. Consensus surveys show that throughput, finality, membership, energy, and fault tolerance involve trade-offs rather than a single ranking [87,88,107].

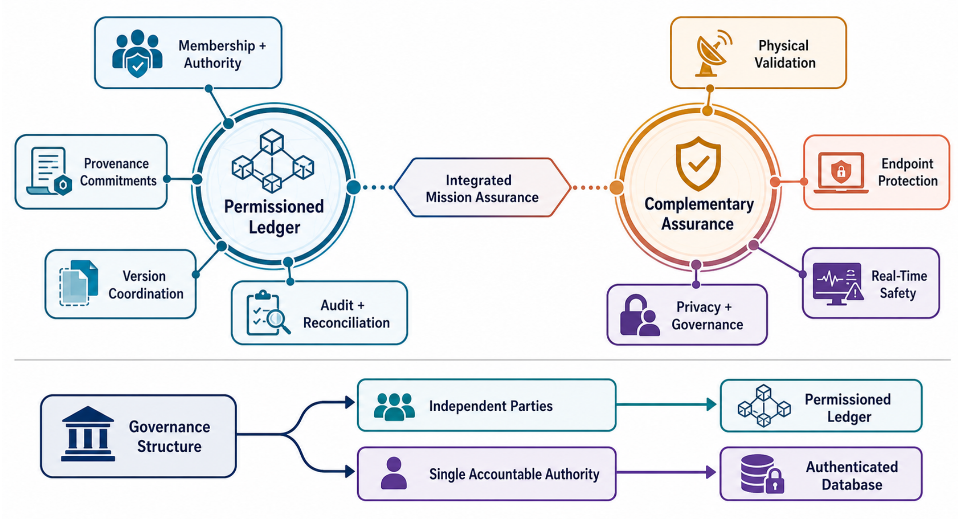

Figure 6. Blockchain boundary in mission trust. The left column identifies suitable ledger functions; the right column identifies complementary assurance responsibilities.

7.2. On-Chain and Off-Chain Partitioning

Raw video, sonar, point clouds, high-rate telemetry, detailed attestation evidence, and fine-grained location should remain off chain, encrypted and subject to data-use policy. The ledger records small, verifiable references: content identifiers or Merkle roots, producer and custodian identifiers, coarse timestamps or time ranges, schema and policy versions, attestation-result commitments, endorsement decisions, revocations, and audit outcomes. The relation is bidirectional. An off-chain object must be verifiably bound to its commitment, and a consumer must be able to determine which ledger state and authorization applied when it was created and used.

On-chain hashes provide integrity commitments. Privacy comes from encryption, access control, selective disclosure, retention rules, key destruction, aggregation, and metadata minimization, because hashes can reveal equality, support dictionary attacks on predictable values, and bind sensitive metadata to a durable member-visible record. On-chain metadata also persists after an off-chain object is deleted. Privacy design therefore accounts for traffic analysis, membership visibility, timing, and the linkage of vehicle identifiers across missions. General IoT and blockchain reviews identify privacy and storage tension [35,36,108]; maritime governance adds commercial and jurisdictional concerns [32,33,109].

7.3. Four Design Patterns

Table 7 organizes recurring ledger uses by OAL claim, assumptions, and complementary assurance requirements. The patterns synthesize UAV, IoT, provenance, and attestation work into reusable design templates.

Table 7. Blockchain design patterns for cross-domain mission assurance.

Pattern	On-chain record	Assurance supported	Necessary assumptions	Complementary assurance required
Membership and authority registry	Organization, role, credential status, delegation and revocation version	Cross-party identity and policy-state consistency	Governed issuer set, protected keys, bounded offline credentials	Real intent, endpoint integrity, emergency authority conflict
Attestation anchor	Evidence digest, verifier identity, result, reference-value and policy version	Accountable device-state appraisal and freshness history	Trustworthy measurement root and verifier; adequate coverage	Sensor truth, unmeasured runtime behavior, compromised verifier
Provenance commitment	Data/version identifier, transformation and custody roots, endorsement	Tamper-evident lineage and accountable processing	Complete instrumentation, secure time and keys, retrievable off-chain evidence	Semantic correctness, biased models, confidentiality of metadata
Partition-aware audit and reconciliation	Signed sequence range, batch root, local policy/model version, reconnect event	Detection of omission, duplication, conflicting logs, and stale authority	Secure local append state, retained evidence, eventual reachability	Missed real-time deadline, events never captured, physical truth
Policy/model coordination	Approved artifact identifier, activation/rollback time, authorized endorsers	Shared knowledge of active rule or model version	Clear governance and deterministic validation of identifiers	Safety and performance of the approved artifact

The membership pattern generalizes UAV identity and authentication proposals [27, 28,31]. It records which organization may issue each role, the applicable policy version, and how offline parties learn of revocation. Short-lived credentials, bounded delegation, and risk-specific offline envelopes constrain exposure while revocation information propagates.

Two full-text authentication studies provide prototype-level evidence for membership and channel establishment. Tan et al. evaluate Hyperledger Fabric smart-contract operations and cryptographic communication/computation costs in a simulated 10 km × 10 km UAV mission with cloud-hosted peers and assumed base-station coverage [28]. SwarmAuth combines PUF-based mutual authentication, a blockchain registry, and location-aware clustering, then evaluates security features, computation and communication cost, throughput, and delay using an EVM implementation and simulations [31]. OAL maps these results to entity/platform admission and ledger-supported registry claims. Observation validity, underwater partition behavior, cross-organizational governance, and consequence-weighted mission decisions form the corresponding cross-domain validation extensions.

The attestation pattern links RATS evidence appraisal to shared audit. An attester produces evidence; a verifier evaluates it under an appraisal policy; a relying party uses a result [41]. Recording a commitment to the result and policy makes later reliance accountable and coordinates verifier outputs among organizations. Attestation architectures for blockchain and edge systems illustrate this intersection [42,43,110,111]. Sensitive measurement detail remains encrypted off chain by default, and the result's validity interval reflects disconnection and device state change.

Heterogeneous attestation is technically plausible, with platform-specific cost. The hardware-agnostic framework of Ott et al. [43] was implemented for TPM, AMD SEV-SNP, and ARM PSA evidence. In that testbed, the attested TLS connection took 31.2 ms with SEV-SNP and 476.1 ms with the tested firmware TPM, the latter dominated by quote generation; report size also depended strongly on representation and trust anchor. These measurements show why an attestation result retains its technology, configuration, freshness, and measurement scope. Mutual attestation supplies software/platform-integrity evidence, complemented by observation evidence for camera alignment, sensor calibration, and physical validity.

The provenance pattern links W3C-compatible derivation to shared commitments. IoT provenance proposals use blockchain to protect provenance records or bind large off-chain collections [112–115]. For mission data, each derived product should identify immutable inputs, transformation code or model, parameters, responsible entity, and output version. Provenance supports explanation and accountability, including for AI artifacts [116]; observation-validity and mission-suitability appraisal establish the semantic quality of the transformation and its output.

The partition-aware pattern is especially relevant to underwater missions. A disconnected node or gateway keeps a signed, monotonically sequenced local log, periodically commits a batch root, and discloses inclusion evidence after reconnecting. Consortium peers validate membership and authority at event time, sequence continuity, duplicates, and policy/model versions. Competing logs can be flagged for governance. This resembles store-carry-forward accountability more than online transaction processing. Under the CAP trade-off, a partition requires a choice between unrestricted availability and immediate global consistency [117]. The mission therefore specifies which local actions remain available and how uncertainty is represented.

7.4. Governance, Revocation, and Recovery

Permissioned consensus places trust in explicit consortium rules. A consortium defines member eligibility, validator distribution, endorsement thresholds, emergency powers, software upgrade, key rotation, data jurisdiction, dispute resolution, incident disclosure, and exit. One-operator governance provides internal tamper resistance and engineering convenience; independently administered validators provide multi-party history. Endorsement thresholds then balance organizational participation with required availability according to the mission and threat model.

Revocation is temporal. A credential revoked at 12:00 may have authorized a disconnected UUV at 11:55; a gateway may learn the change at 12:30. Audit needs event time, receipt time, local policy version, credential validity evidence, and reconciliation time. Recovery covers identities, affected models, data products, provenance descendants, device re-attestation, and review of consequential mission decisions. A ledger preserves the dependency identifiers needed to compute this affected set, while the OAL graph guides the complete recovery scope.

7.5. Real-Time and Safety Boundaries

Hard control loops use local verified control and deterministic timing for vehicle stabilization, collision avoidance, emergency surfacing, and other deadline-bound safety functions. The ledger distributes approved policy or software versions before operation, records summaries afterward, and supports slower supervisory decisions, cross-party authorization, data release, and after-action audit. This division preserves fundamental platform safety during ledger-service loss while making reduced audit freshness and cross-organizational coordination visible.

Figure 6 provides a claim discipline. Blockchain strengthens evidence about who agreed to what, under which version, and in which order. Physical, endpoint, timing, privacy, and governance evidence completes the assurance case for whether the world matched the record, endpoints were secure, connectivity and deadlines met mission needs, and consortium rules were legitimate.

8. Evaluation Practices and Evidence Maturity

The literature spans conceptual architectures, analytical models, simulations, prototypes, datasets, and a smaller number of domain demonstrations. Each form answers a distinct question: security analysis exposes protocol assumptions; network simulation explores scale; blockchain prototypes measure implementation behavior; and field trials demonstrate integration under representative conditions. Integrated mission-trust claims combine these forms so that acoustic realism, organizational governance, sensor validity, adversarial behavior, and system performance are evaluated at the appropriate level.

8.1. An Evidence-Maturity Ladder

Figure 7 proposes a six-level ladder for claims about cross-domain mission trust. The ladder orders ecological evidence while preserving the distinct value of each method: strong mathematical analysis at level 1 can be more informative about a security property than a poorly controlled field demonstration at level 4. Advancement exposes and tests additional classes of assumptions. Claim language is matched to the level reached.

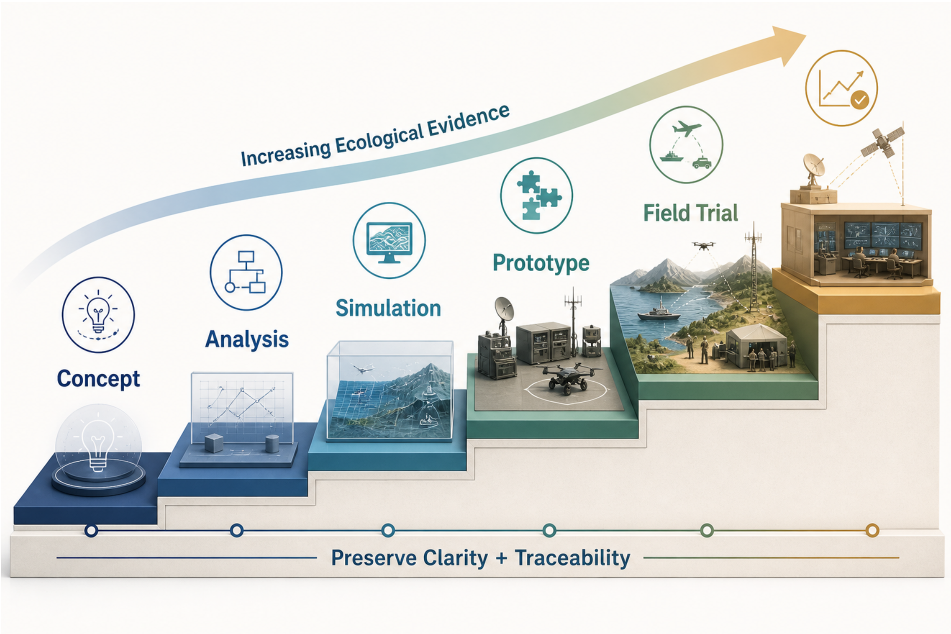

Figure 7. Authors' evidence-maturity ladder. Progress requires preserving lower-level clarity while adding implementation, environmental, organizational, and longitudinal evidence.

At level 0, a work offers a use case, taxonomy, or architecture with traceable reasoning and establishes conceptual value. Level 1 adds mathematical, formal, or security analysis with explicit adversary and system assumptions. Level 2 uses simulation or recorded data to test behavior under controlled variation. Level 3 integrates an implementation in a prototype or testbed and reports configurations, code, and workload. Level 4 conducts trials in representative air, surface, or underwater conditions with realistic communications and operational participants. Level 5 demonstrates sustained multi-party operation, including governance changes, credential and software rotation, failures, audits, and recovery. A ledger-throughput test supplies level-3 evidence for the ledger component; a complete mission-trust claim advances through corresponding observation and mission-use tests.

8.2. Datasets, Simulators, and Testbeds

Existing public resources form a modular validation stack for UAV–USV–UUV motion, heterogeneous links, device attestation, provenance, adaptive trust, consortium governance, and mission outcomes. TON_IoT combines labeled IoT/IIoT telemetry, operating-system logs, and network traffic collected in the UNSW Canberra cyber range; Edge-IIoTset uses a purpose-built seven-layer testbed and supplies baselines for centralized and federated learning [118,119]. Together with CICIoT2023 [120], they support device/link anomaly experiments and can be paired with ocean-physics, vehicle-dependency, and mission-loss modules. SeaDronesSee provides real maritime UAV imagery and metadata for observation-validity experiments [121]. The supervised GNSS study of Semanjski et al. [65] validates on two distinct real-world spoofing/meaconing datasets, contributing signal-specific evidence from real-world experimental contexts.

AirSim combines a high-frequency physics engine, realistic visual environments, and real-time hardware-in-the-loop support, with selected software components compared against real flights [122]. UUV Simulator extends Gazebo/ROS with hydrodynamics, thrusters, sensors, external disturbances, and multi-robot scenarios [123], while Aqua-Sim NG ports underwater-network functionality to ns-3 and evaluates the redesigned simulator architecture [124]. These resources contribute complementary evidence: AirSim emphasizes air-domain perception and dynamics, UUV Simulator physical robot integration, and Aqua-Sim NG communication protocols. Timestamped event interfaces can couple them, with time synchronization, shared geometry, channel abstraction, reproducibility, and interface validation reported as part of the experiment.

Table 8. Evaluation resources, the OAL cells they can exercise, and transfer limits.

Resource/practice	Primary evidence	OAL use	Important metrics	Required complement
CICIoT2023, TON_IoT, Edge-IIoTset	Labeled network, host, sensor, and attack telemetry; centralized/federated baselines	Device/link anomaly and uncertainty methods	Detection by class, calibration, false isolation, delay, missingness sensitivity	Cross-media maritime topology, physics, and mission-loss model
SeaDronesSee and GNSS records	Real maritime imagery/metadata and real-world signal-manipulation records	Observation validity, data-product quality, spoofing context	Detection error, geolocation error, uncertainty calibration, robustness to weather/viewpoint	Underwater, ledger, and additional observation contexts
AirSim	High-frequency physics, sensors, visual worlds, hardware-in-the-loop	Air-domain observation and mission use	Mission success, safety constraint violations, sensor/model shift	Acoustic-network and consortium-governance modules
UUV Simulator	ROS/Gazebo hydrodynamics, thrusters, sensors, disturbances, multi-robot scenarios	Underwater motion, sensing, control, integration	Localization, energy, mission completion, failure recovery	Field validation of hydrodynamics and sensor behavior
Aqua-Sim NG/ns-3	Underwater-network protocols, channel abstractions, routing experiments	Link/path evidence, delay, loss, routing and partition	Delivery, latency distribution, energy, path churn, outage duration	Physical observation-validity module
Permissioned-ledger testbed	Peers, ordering, endorsement, contracts, failures	Ledger/governance consistency and audit	Commit/finality latency, throughput, resource use, recovery, fork/conflict handling	Representative evidence sizes, event rates, and duty cycles
Cross-domain field exercise	Physical platforms, operators, weather, real links	Integrated lifecycle and mission-use suitability	Consequence-weighted mission outcomes, false quarantine, recovery time, audit completeness	Repeatable scenarios and systematic adversarial coverage

8.3. Metrics that Respect the Assurance Chain

Evaluation should report at least five metric families. First, *inference quality*: discrimination where labels exist, calibration or interval coverage, uncertainty under missing evidence, detection and recovery delay, and performance under distribution shift. Accuracy alone is inadequate because a rare false isolation can remove a critical relay. Second, *mission effect*: task completion, time, coverage, safety-constraint violations, decision regret, and resource allocation under adversarial and benign degradation. Third, *network cost*: bytes per evidence event, delay distribution rather than only mean delay, energy, storage growth, duty cycle, and partition duration. Fourth, *ledger behavior*: endorsement and finality latency, throughput under realistic event batching, validator/resource cost, membership change, outage behavior, reconciliation completeness, and disputed histories. Fifth, *accountability*: percentage of decisions with resolvable provenance, time to determine affected descendants, policy/model version traceability, and successful revocation and recovery.

Experimental factors should cross object, threat, and context dimensions. Examples include compromised entities versus devices; isolated versus colluding nodes; jamming versus environmental fading; fresh versus stale attestation; independent versus correlated sensors; honest versus compromised gateways; short versus prolonged partitions; and routine versus high-consequence mission policies. A factorial design can reveal interactions that single-attack demonstrations miss. Crucially, ground truth must distinguish malicious action, benign degradation, and missing evidence. Otherwise a detector can appear successful by learning weather or topology artifacts.

Baselines should also be layered. A centralized authenticated database measures blockchain's added value; a signed local log measures the contribution of replication; static access control measures the value of continuous evidence; object-local trust measures graph propagation; a no-provenance condition measures lineage; and a no-uncertainty condition reveals overconfident decisions. Blockchain comparisons should hold endorsement, workload, hardware, and fault assumptions constant. Studies of Fabric-based access control and integrated IoT ledgers offer implementation starting points [112,125], completed by an application-specific baseline.

Reproducibility requires the exact topology, channel and mobility models, workload, cryptographic settings, ledger release and configuration, endorsement and fault model, random seeds, data partitions, trust-policy versions, and mission-loss function. Versioned provenance should cover the evaluation itself. This is more than archival hygiene: if a policy threshold or model changes between runs, the reported mission outcome belongs to a different data product and assurance claim.

8.4. Interpreting Claims by Evidence Level

UAV blockchain applications range from security and privacy architectures to swarm monitoring and resilient-network proposals [94,126–129]. Their evidence maps to the corresponding level of the maturity ladder. Broader work on Industry 4.0, federated learning, digital twins, supply chains, and government sharing contributes reusable organizational patterns [130–134]. Cross-domain validation then combines these patterns with endpoint, link, timing, governance, and mission-consequence evidence under intermittent connectivity.

The supplementary matrix supports this discipline by linking DOI-verified records, lawful full-text acquisition, and claim-level analytical coding. The priority acquisition set is complete, and the studies used for method-specific comparisons in Sections 6–8 have been coded from full text. The remaining screened records support discovery and thematic context and provide a transparent roadmap for subsequent claim-level coding.

9. Discussion and Research Agenda

9.1. Theoretical Implications

OAL reframes trust from a property attached to a vehicle into a structured relying relation. This has three theoretical consequences. First, trust is compositional through typed dependency relations and explicit inference rules. Confidence in an entity supports authorization, device attestation supports execution state, and observation and mission evidence complete the downstream claims. Second, trust is purpose-bound. Evidence sufficient for sharing low-sensitivity imagery may be insufficient for cooperative navigation, even when the underlying object is identical. Third, uncertainty is an operational state: "Unknown" maps to a documented policy response that preserves the distinction from trusted and malicious states.

The assurance chain also clarifies the relationship between provenance and truth. Provenance is an argument about history: who generated an artifact, through which activities, from which inputs. Observation validity is an argument about the world. Ledger consistency is an argument about shared records. Mission suitability is an argument about consequence. These arguments can reinforce each other but remain falsifiable independently. Provenance and decision-accountability research supports the value of traceable transformations [104,116]; OAL adds the requirement that a mission consumer name which history is sufficient for which use.

9.2. System Resilience under Partition and Degradation

Future architectures should treat partition as a routine operating mode. Local policy envelopes require bounded capability, expiry, evidence-age limits, safe fallback, and explicit record of which global facts were unavailable. Reconnection combines data synchronization with semantic reconciliation: the system decides whether actions were authorized at event time, whether a newly received revocation invalidates descendants, and whether conflicting observations require retrospective mission review. Blockchain scalability research focuses on throughput and storage [135,136]; cross-domain missions add long disconnection, sparse contacts, and consequence-sensitive local autonomy.

Resilience also requires functional diversity. If the USV is simultaneously acoustic gateway, time source, fusion node, policy-enforcement point, and validator, redundant vehicle count can conceal a single logical point of failure. The dependency graph can expose this concentration by measuring which vertices support many mission paths. Future work should compare architectural redundancy (multiple instances) with evidential independence (different roots, media, sensors, and organizations). Two reports relayed through the same compromised gateway share a common cause and therefore count as one evidential lineage.

9.3. Governance, Privacy, and Human Authority

Consortium governance is part of the security mechanism. Research should specify who can admit validators, approve reference values, activate contracts and models, invoke emergency powers, decide disputes, and audit the auditors. Maritime blockchain adoption studies already identify policy and organizational factors beyond technical performance [32,34,109]. In mission assurance, a governance failure can propagate like a cyberattack: an unsafe reference value makes compromised devices appear healthy, or a delayed revocation keeps stale authority active.

Privacy questions extend beyond encrypting payloads. Cross-mission identifiers, validator endorsements, timing, route custody, and model-use records can reveal organizational relationships or vehicle activity. Work on blockchain and IoT warns of immutable metadata and access-control tension [36,72,137]. Future designs should evaluate unlinkability, data minimization, selective disclosure, authorization-policy confidentiality, and deletion semantics alongside audit value. A mission may need accountable pseudonymity: an operator can verify that an authorized role acted without learning the long-term identity, while a governed process can resolve accountability after an incident.

Human authority must remain explicit. Automated trust inference can recommend reduced privilege, corroboration, or isolation, while high-consequence recovery and dispute decisions may require human review. The relevant evidence comprises model output, explanation, policy version, uncertainty, alternatives, and consequence. Authorized and auditable human override preserves this evidence trail and the accountability of the relying decision.

9.4. Testable Research Propositions

The framework yields propositions that future studies can falsify.

P1 (assurance separation). Systems that expose separate observation, provenance, ledger, and mission-suitability states will reduce unsafe acceptance of correctly signed but semantically invalid data compared with systems using a single node-reputation score, under matched false-isolation cost.

P2 (dependency awareness). Trust decisions that preserve evidence lineage and shared-cause dependencies will be better calibrated under correlated sensors and colluding recommenders than decisions that aggregate object scores independently.

P3 (three-way diagnosis). Policies that distinguish malicious, benign-degraded, and unknown states will achieve lower mission loss during mixed environmental and adversarial conditions than binary trusted/untrusted policies.

P4 (bounded offline authority). Time-, resource-, and consequence-bounded offline authorization will maintain higher mission availability than fail-closed global authorization while producing lower retrospective policy violation than unrestricted offline autonomy.

P5 (selective ledger value). A permissioned ledger will improve cross-organizational audit completeness and dispute resolution relative to signed centralized logs when governance is genuinely distributed, but will not improve hard real-time control performance and may degrade it if placed in the control path.

P6 (maturity gap). Designs evaluated only through transaction or network metrics will overstate mission assurance relative to evaluations that combine observation validity, endpoint compromise, path degradation, governance change, and consequence-weighted mission outcomes.

Each proposition supplies a comparative, refutable design for future empirical work and requires operational definitions and preregistered decision losses.

9.5. From OAL Cells to an Assurance Case

OAL becomes operational when each high-consequence mission use is expressed as a compact assurance case. The top claim should be deliberately narrow: for example, "fused product d_{17} is suitable for route replanning by consumer c until time t under weather class w and policy p." The argument then descends through the four layers. Observation subclaims identify the sensors, calibration, geometry, environmental model, residuals, and corroboration that justify physical plausibility. Commitment subclaims identify immutable inputs, producers, transformations, custody, time, and protected versions. Ledger subclaims state the membership, endorsement, ordering, and policy state shared by organizations. Mission subclaims connect freshness and uncertainty to an explicit consequence and permitted action. Any unsupported subclaim is displayed as a defeater, caveat, or reason to restrict action.

This structure records both supported scope and boundary conditions. A verifier may establish that a measured software component matches an approved reference, with camera alignment and operator intent assigned to separate subclaims. A provenance record may establish the instrumented derivation path and identify a legacy transformation as a gap in completeness. A ledger may establish endorsement of a model version by three authorized peers, while safety and accuracy are supported by separate validation evidence. Pairing each guarantee with its conditions gives downstream consumers the exact scope of justified reliance.

Assurance cases must also be time-aware. Evidence has an observation time, issue time, receipt time, and expiry condition; a dependency can change between them. If a USV attestation is fresh when it receives a UUV batch but stale when it transforms and endorses that batch, the later activity requires an explicit continuity assumption or renewed evidence. Clock uncertainty is represented where GNSS is unavailable or suspected. For disconnected operation, the case records the last globally known policy and the local authorization envelope, preserving the scope of agreement available at event time.

Counter-evidence is first-class. A new sensor residual, revocation, incident report, or conflicting custody event should identify which claims and descendants become disputed. This enables proportional response: quarantine a derived track while preserving independently supported observations from a platform, or reduce a gateway's endorsement role while retaining its emergency relay function. Recovery closes the argument through review of affected credentials, software, models, data descendants, and mission decisions, producing a scoped and evidence-backed return to service.

Future tools could generate these arguments from the typed dependency graph while preserving human judgment. Policy authors decide evidence sufficiency, independence, consequence, and acceptable uncertainty. The generated artifact therefore distinguishes machine-verified bindings (signature, hash, sequence), verifier appraisals (measured state against reference), statistical assessments (calibration or anomaly), and governance judgments (waiver, dispute, emergency authority). A reviewer can then compare the conclusion with the strongest supported subclaim.

Finally, evaluation should measure the assurance case itself. Completeness can be assessed as the fraction of consequence-relevant dependencies with current evidence; soundness can be probed through seeded counterexamples; usability can be measured by the time and error with which operators identify affected products; and governance quality can be tested through conflicting but authorized decisions. These measures complement detection and ledger performance. They ask whether the architecture supports defensible reliance, which is the conceptual goal of mission trust.

9.6. Toward a Cross-Domain Unmanned Trust Profile

The proposed Cross-Domain Unmanned Trust Profile (CDUTP) is an interoperability profile defining a small, versioned set of claims and evidence references for the OAL cells most often crossed by organizations. NIST Zero Trust Architecture supplies resource- and request-centered policy principles [39]; IETF RATS supplies attestation roles and semantics [41]; W3C PROV-DM and PROV-O supply entity–activity–agent provenance relations [44,45]. NIST's IoT device baseline informs minimum endpoint capabilities [68]. 3GPP specifications for UAS support and maritime services provide communications and service context [138,139]. Permissioned-ledger governance and endorsement supply multi-party state coordination [105,106].

Software and model update evidence reuses secure-update architecture. The IETF SUIT architecture separates update manifests, authorization, payload acquisition, installation, and device constraints [140]; CDUTP could reference such an update identifier and result while keeping the firmware payload off chain. The profile thereby binds existing domain claims into a mission assurance graph through their established protocols.

A profile instance should minimally state: object identifier and type; claim and assurance layer; issuer and relying role; evidence or provenance reference; observed, issued, and expiry times; mission context and permitted use; uncertainty or appraisal outcome; applicable policy, model, schema, and contract versions; privacy label and disclosure conditions; parent dependencies; revocation reference; and on/off-chain binding. It should support disconnected issuance and later verification while labeling freshness at the relying time. Compact encodings are desirable for acoustic links, but semantic equivalence is more important than one wire format.

Standardization should proceed in stages. First, publish use cases and claim-scope requirements. Second, define a shared vocabulary and conformance tests for evidence lineage and version binding. Third, run plugfests across independently implemented UAV, USV, UUV, gateway, verifier, and ledger components. Fourth, evaluate privacy and governance with multiple organizations, including member exit and disputed recovery. Fifth, pursue alignment with relevant standards bodies. CDUTP is proposed here as the authors' research agenda for that future alignment.

9.7. Design Heuristics

Several practical heuristics follow. Begin with the relying decision and select the ledger role to match it. Name the object and assurance claim before choosing a trust method. Preserve evidence identity and common causes. Keep raw evidence off chain and minimize metadata. Make staleness and unknown state visible. Separate hard real-time safety from shared audit. Treat policies, models, contracts, and reference values as versioned, threat-exposed artifacts. Test a centralized signed-log baseline. Finally, match language to maturity: "proposed," "analyzed," "simulated," "prototyped," "field-tested," and "deployed" denote distinct evidence levels.

These heuristics yield direct technology-selection criteria. Signed append-only logs and replicated databases suit settings with one accountable manager of state and audit. Consensus adds value when independent parties require a common history. Ledger commitments require platforms that can maintain keys, time, and protected local state for evidence binding. Decisions with deadlines shorter than end-to-end communication rely on local safety assurance.

Responsible deployment adds one final boundary. Mission-trust infrastructure can improve safety, environmental observation, inspection, disaster response, and coalition accountability, while persistent identities and provenance traces also create surveillance and operational-security consequences. An assurance case should therefore include a legitimate-purpose claim, proportional data collection, retention and disclosure limits, and an accountable path for human appeal or incident review. Dual-use assessment focuses on capabilities and governance. Public reporting can disclose architecture, assurance semantics, failure modes, and evaluation methods while protecting credentials, exact operational configurations, exploitable topology details, and mission-specific tactics. This balance supports scientific testing and responsible dissemination. It also reinforces the core OAL insight: governance artifacts and privacy policies are versioned, contestable evidence within the mission system and require the same accountable appraisal as technical artifacts.

10. Limitations of the Review and Framework

This review uses a narrative design and a focused English-language evidence corpus with a principal 2016–2026 window. The design supports conceptual synthesis across control, robotics, communications, cybersecurity, distributed systems, provenance, trust modeling, and governance, while leaving source discovery, grouping, and emphasis partly dependent on author judgment. The evidence matrix makes those choices traceable. Systematic estimates of literature prevalence would require a separately designed multilingual search, screening protocol, and sampling frame.

Lawful full texts were obtained for all 33 sources in the priority acquisition set, and 16 studies received claim-level coding for the method-specific comparisons in this version. The remaining screened records provide thematic orientation and a transparent roadmap for extending full-text analytical coding across the 140-source corpus. Detailed performance and comparative statements are linked to completed coding, and future revisions can deepen the matrix as additional configurations, assumptions, and limitations become analytically relevant.

Much evidence is transferred from adjacent domains. UAV security, IoT reputation, road-vehicle trust, industrial provenance, and permissioned enterprise ledgers provide mechanisms whose threat distributions, time scales, connectivity, energy, and governance are re-evaluated for air–surface–underwater operations. Even within maritime systems, acoustic conditions and operational practices vary widely. The OAL framework makes

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these transfer assumptions visible and turns them into target-environment validation requirements.

OAL, the dependency relation set, CTG loop, maturity ladder, propositions, and CDUTP form the authors' conceptual architecture and define a staged validation agenda. Integrated evaluation should use mission data, formal property analysis, interoperable implementations, and standards-oriented conformance work. Applications involving human teams, biological sensors, digital twins, or legal jurisdictions may refine the four object types. Practical graph implementations will also require rules for abstraction, correlation, privacy, evidence expiry, and cycle handling. Empirical calibration determines when the general notation $T_o(t, c, m)$ can be interpreted probabilistically.

Finally, the public-evidence focus supports reproducible analysis of assurance, resilience, audit, and responsible governance in civil and publicly documented defense-support settings. Sensitive operational details remain within the governance and disclosure controls of the organizations that hold them. Future authorized studies can evaluate the framework under additional contested-environment and platform-specific conditions.

11. Conclusions

Air–surface–underwater cooperation turns trust into an end-to-end mission problem. A defensible decision integrates a plausible observation, an intact and accountable provenance chain, a correctly governed shared state where one is needed, and evidence that remains suitable for the particular mission consequence. The authors' OAL framework connects these four assurance layers to entity, platform/device, link/path, and data-product objects over the complete lifecycle from admission to recovery.

The object-dependency graph and CTG loop show how evidence, uncertainty, and attacks move across that structure. They preserve malicious, benign-degraded, and unknown states according to the available support, and they associate numerical trust scores with explicit calibration evidence. The evidence-maturity ladder aligns published claim language with the demonstrated level of analysis, implementation, environmental realism, and operational experience.

Blockchain is most valuable where independent organizations need consistent membership, attestation and provenance commitments, version coordination, accountable audit, and post-partition reconciliation. Physical corroboration establishes sensor truth; attestation and runtime controls protect endpoints; local verified control handles partitions and hard deadlines; encryption and minimization provide privacy; and explicit consortium rules provide governance. A conventional authenticated system remains the efficient choice where one accountable manager governs state and shared audit has limited value.

The proposed CDUTP and six research propositions provide a route from conceptual synthesis to interoperable, testable systems. The priority evidence audit gives those propositions a firmer literature basis, and the next stage is empirical validation through coupled cross-domain testbeds, representative field trials, and multi-party governance exercises. Success is measured by whether every consequential mission use can be traced to appropriate, current, and explicitly scoped assurance.

Supplementary Materials: The following supporting information accompanies the submission: Table S1, the internal evidence matrix (CSV); Table S2, the lawful full-text acquisition manifest (CSV); and the publication figure files for Figures 1–7 (PDF).

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Abbreviations

AUV	Autonomous underwater vehicle
CDUTP	Cross-Domain Unmanned Trust Profile
CTG	Context–temporal–graph
DLT	Distributed ledger technology
OAL	Object–Assurance–Lifecycle
RATS	Remote ATtestation procedureS
UAS	Unmanned aerial system
UAV	Unmanned aerial vehicle
USV	Unmanned surface vehicle
UUV	Unmanned underwater vehicle
ZTA	Zero trust architecture

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